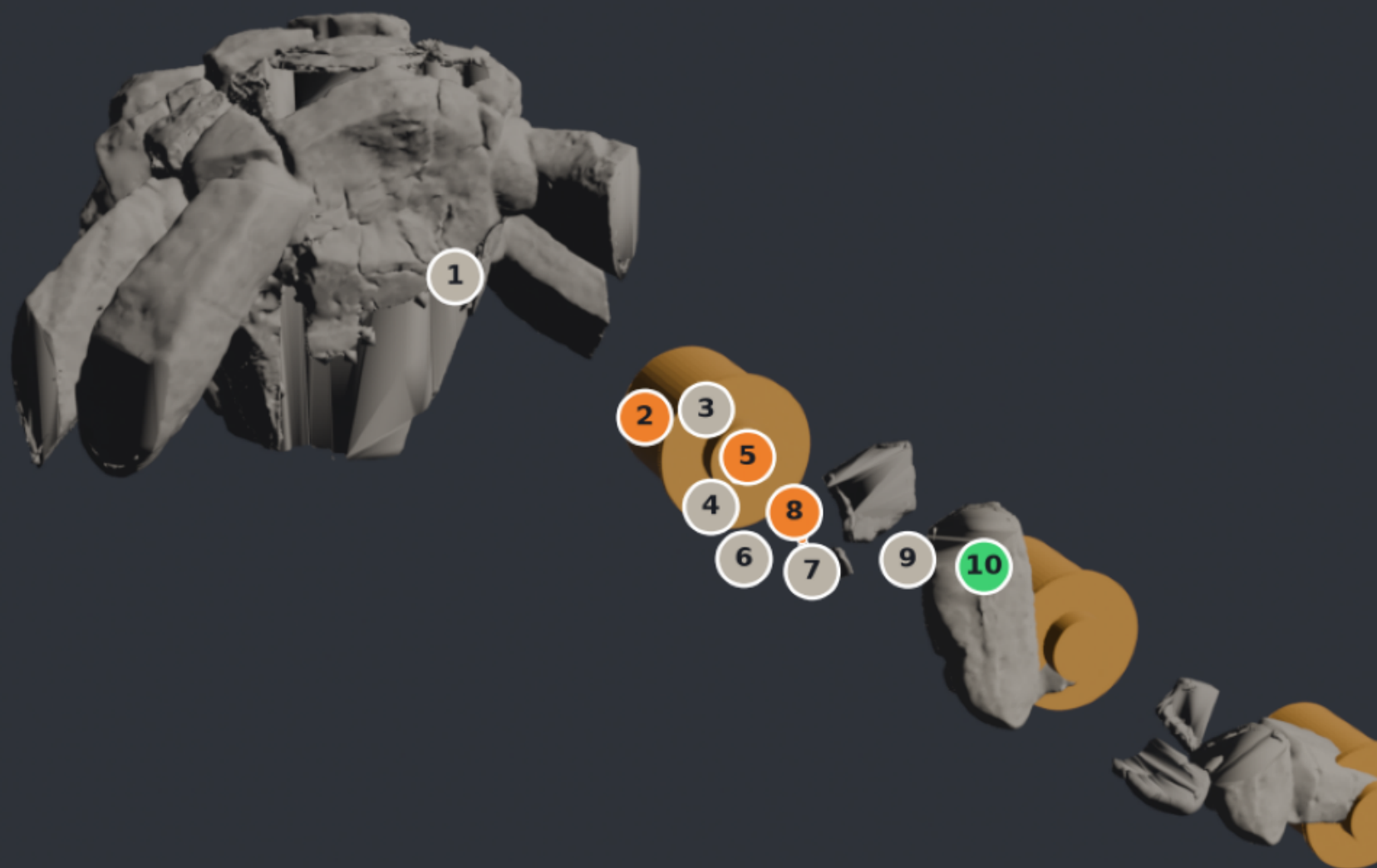


ROCKY-5 — Exploded Leg

One leg along its axis: thorax socket → hip servo → coxa → knee1 servo → femur → knee2 servo → tibia → hand. Assemble proximal → distal.



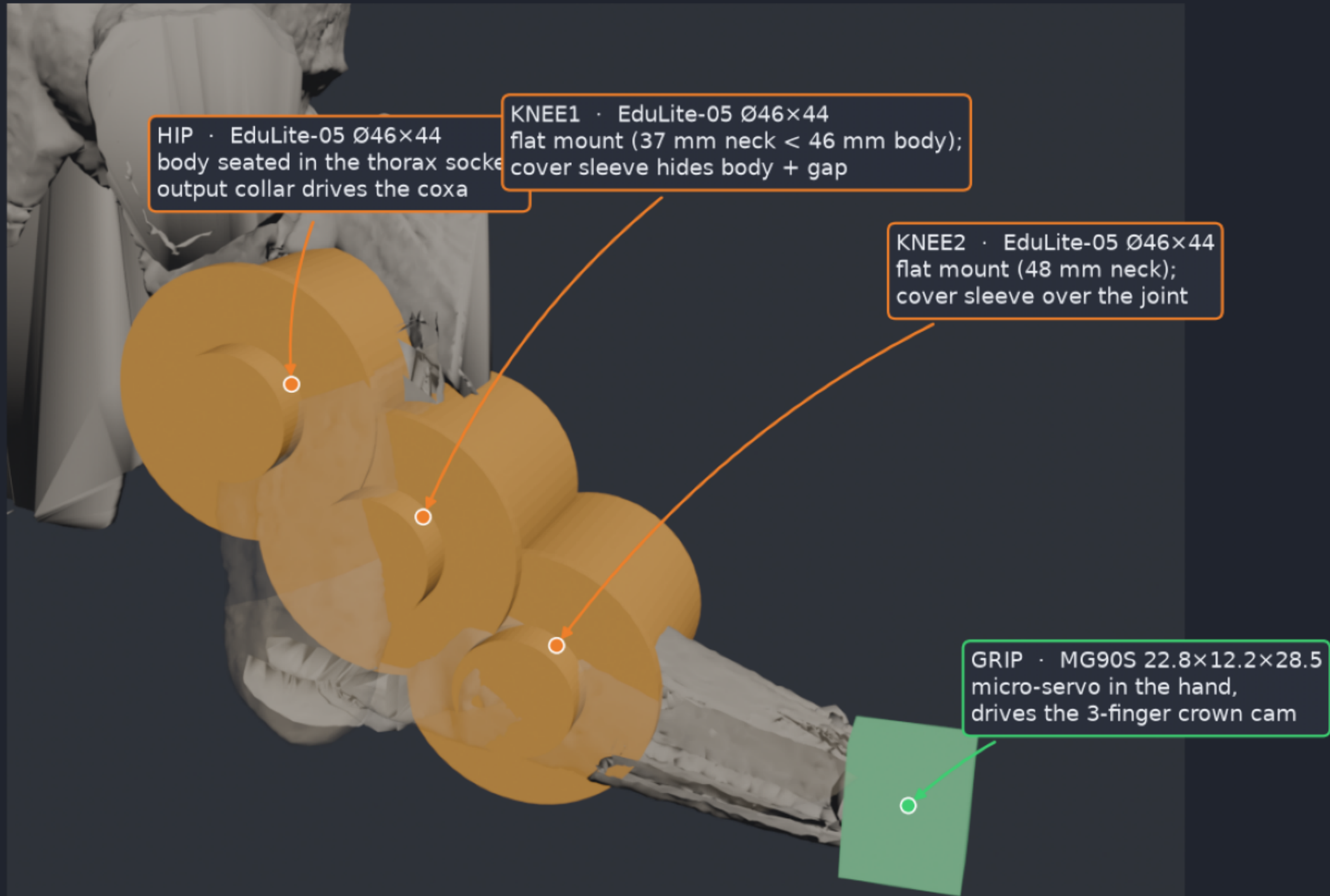
ASSEMBLY ORDER

- 1 Thorax dome**
hosts the hip socket (Ø46 pilot); leg root sinks in
- 2 HIP servo**
EduLite-05 Ø46×44 — buried in the thorax socket
- 3 Coxa**
28.8×38.3×47.7 mm shell (×5)
- 4 Knee1 cover**
cosmetic stone sleeve — clips over servo (×5)
- 5 KNEE1 servo**
EduLite-05 Ø46×44 — flat mount (37 mm neck)
- 6 Femur**
33.1×51.9×58.6 mm shell (×5)
- 7 Knee2 cover**
cosmetic stone sleeve — clips over servo (×5)
- 8 KNEE2 servo**
EduLite-05 Ø46×44 — flat mount (48 mm neck)
- 9 Tibia + hand**
74.5×31.3×83.8 mm; sacred 3-finger hand distal (×5)
- 10 GRIP micro**
MG90S 22.8×12.2×28.5 — drives the 3 fingers

Orange = EduLite-05 leg servo Green = grip micro
Stone = printed shell / cover

ROCKY-5 — Leg Cutaway

Near shell wall clipped away: each EduLite-05 body sits in the proximal socket, its Ø24 output collar drives the next segment. Now the joint reads as a mechanism.

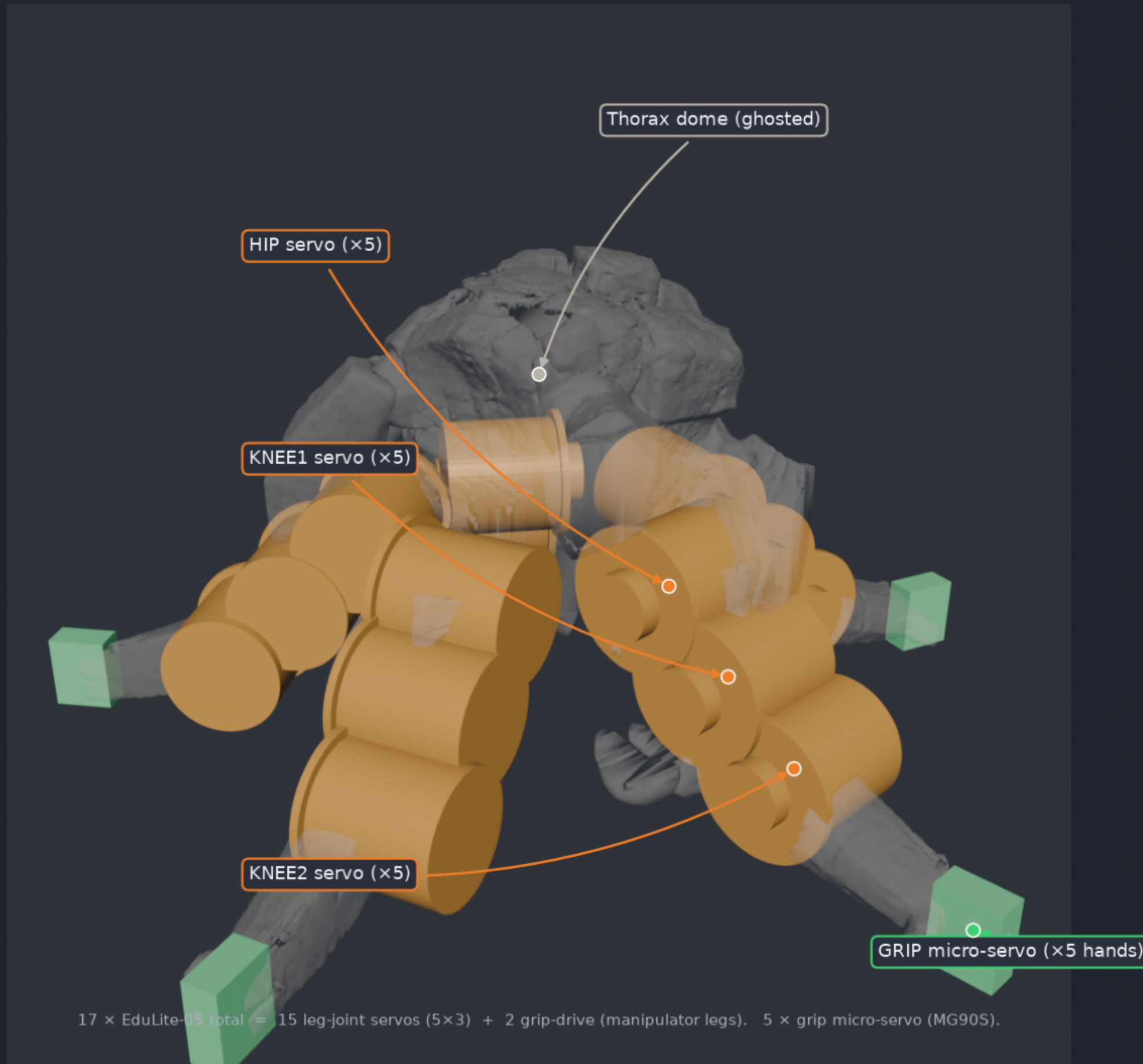


EduLite-05 = Ø46×44 housing + Ø24×8 output collar + Ø52 mounting flange (Ø41.5 PCD).

The 25 mm coxa↔femur gap at knee1 is the servo seat pocket — the modelled Ø46 body fills it (coxa 10.9 mm & femur 11.8 mm from the servo axis → both inside the 23 mm body radius: BRIDGED).

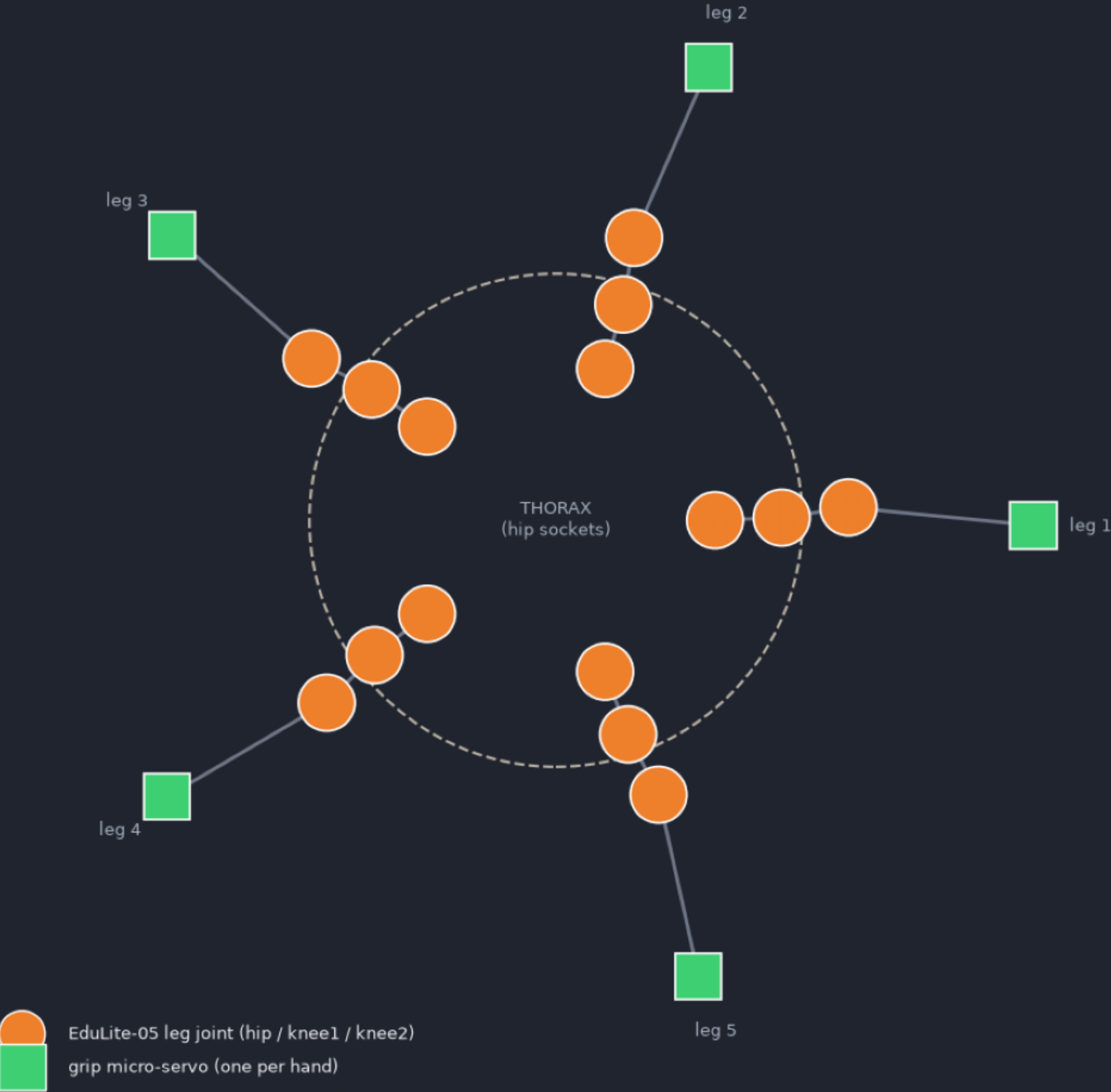
ROCKY-5 — Ghosted Assembly

Segments semi-transparent so every servo is visible in place: 3 EduLite-05 per leg $\times$ 5 legs, plus a grip micro-servo in each hand.



ROCKY-5 — Servo Layout & BOM

Top-down schematic of all servo positions (true 5-fold radial symmetry) with the actuator bill of materials.



ACTUATOR BOM

QTY	SERVO · SIZE · ROLE
15	EduLite-05 Ø46×44, 242 g leg joints: 5 legs × (hip, knee1, knee2)
2	EduLite-05 Ø46×44, 242 g grip drive (manipulator legs 1 & 4)
= 17	EduLite-05 total Ø41.5 PCD, Ø24 output 1.8 / 6 N·m QDD actuator
5	grip micro MG90S 22.8×12.2×28.5, 13.4 g 3-finger hand — one per hand
1	MG90S (breathing) 22.8×12.2×28.5 carapace scroll-cam (not a joint)

EduLite-05 total: 17
Grip micro total: 5 (+1 breathing-cam micro)
Source: common/cad_lib/components.rocky_components()